

7th Grade Unit 6 Assessment Guide

General Information	The assessment covers three benchmarks, MA.7.AR.3.2, MA.7.AR.3.3, and MA.7.AR.4.5. Of these three benchmarks, MA.7.AR.3.2 and MA.7.AR.4.5 are similar to each other. The test item specifications state that MA.7.AR.3.2 and MA.7.AR.4.5 will be assessed together. This grouping indicates the similarity between these benchmarks. Encouraging students to correctly show units during instruction is important but this assessment does not rely on the student providing the unit as the units are provided in the table or answer space. Teachers may choose to edit the assessment to require students to give the correct unit. A common error that students make is an incorrect set-up of a proportion. If students do this throughout the assessment, consider working with these students in a small group to determine the reason for their misconception and to provide the student a chance to show mastery after remediation. Sample work for the answers is provided but may not be the way students show their work. It is provided to support teachers.
Calculator Information	Students should have access to scientific calculators for this assessment.
Reference Sheet	Students should have access to the conversions on the Grade 7 B.E.S.T. Mathematics Reference Sheet.
Total Recommended Points	The total recommended points in this guide are 96 points. Teachers can choose to make the total number of points 100 by changing the number of points for different questions.

Question	1a	Benchmark(s)	MA.7.AR.3.3
$2500 \text{ Yen} \cdot \frac{\$1}{267.38 \text{ Yen}} = \9.35	Recommended Scoring (8 Total Points)	Consider giving partial credit for students who correctly show work.	
	Additional Notes	Student work will vary. The dollar sign is not necessary since the table notes US dollars.	
Question	1b	Benchmark(s)	MA.7.AR.3.3
$\$99 \cdot \frac{84.90 \text{ Yen}}{\$1} = 8405.10$	Recommended Scoring (8 Total Points)	Consider giving partial credit for students who correctly show work.	
	Additional Notes	Student work will vary. Students may write the answer as 8405.1 since students may not be familiar with the Yen currency.	

Question	2	Benchmark(s)	MA.7.AR.3.3
$7 \text{ oz} \cdot \frac{28.35 \text{ g}}{1 \text{ oz}} = 198.45$ OR $7 \text{ oz} \cdot \frac{1 \text{ g}}{0.0352 \text{ oz}} \approx 198.86$		Recommended Scoring (8 Total Points)	Consider giving partial credit for students who correctly show work.
		Additional Notes	Students may arrive at slightly different answers depending on which conversion they use.
Question	3 (length)	Benchmark(s)	MA.7.AR.3.3
$350 \text{ in.} \cdot \frac{2.54 \text{ cm}}{1 \text{ in.}} \cdot \frac{1 \text{ m}}{100 \text{ cm}} = 8.89$ OR $350 \text{ in.} \cdot \frac{1 \text{ cm}}{0.39 \text{ in.}} \cdot \frac{1 \text{ m}}{100 \text{ cm}} \approx 8.974$		Recommended Scoring (8 Total Points)	Consider giving partial credit for students who correctly show work. Students may choose to show their work as two steps.
		Additional Notes	Students may arrive at slightly different answers depending on which conversion they use.

Question	3 (width)	Benchmark(s)	MA.7.AR.3.3
$90 \text{ in.} \cdot \frac{2.54 \text{ cm}}{1 \text{ in.}} \cdot \frac{1 \text{ m}}{100 \text{ cm}} = 2.286$ OR $90 \text{ in.} \cdot \frac{1 \text{ cm}}{0.39 \text{ in.}} \cdot \frac{1 \text{ m}}{100 \text{ cm}} \approx 2.308$		Recommended Scoring (8 Total Points)	Consider giving partial credit for students who correctly show work. Students may choose to show their work as two steps.
		Additional Notes	Students may arrive at slightly different answers depending on which conversion they use.
Question	4	Benchmark(s)	MA.7.AR.3.3
$5 \text{ cups} \cdot \frac{0.24 \text{ liters}}{1 \text{ cup}} \cdot \frac{1000 \text{ mL}}{1 \text{ liter}} = 1200$ OR $5 \text{ cups} \cdot \frac{1 \text{ liter}}{4.23 \text{ cups}} \cdot \frac{1000 \text{ mL}}{1 \text{ liter}} \approx 1182.033$		Recommended Scoring (8 Total Points)	Consider giving partial credit for students who correctly show work. Students may choose to show their work as two steps.
Question	5	Benchmark(s)	MA.7.AR.3.2, MA.7.AR.4.5
$100000 \text{ km} \cdot \frac{1 \text{ mile}}{1.61 \text{ km}} \approx 62111.8$ OR $100000 \text{ km} \cdot \frac{0.62 \text{ mile}}{1 \text{ km}} = 62000$		Recommended Scoring (8 Total Points)	Consider giving partial credit for students who correctly show work.

Question	6	Benchmark(s)	MA.7.AR.3.2, MA.7.AR.4.5
$\frac{1}{5} = \frac{x \text{ ounces of sugar}}{27 \text{ ounces}}$ 5.4 ounces of sugar		Recommended Scoring (8 Total Points)	Consider giving partial credit for students who correctly show work.
		Additional Notes	Students may choose to solve the problem using a variety of methods that indicate an understanding of the proportional relationship as a unit rate.
Question	7	Benchmark(s)	MA.7.AR.3.2, MA.7.AR.4.5
$\frac{56 \text{ days}}{500 \text{ miles}} = \frac{x \text{ days traveling along the river}}{25.3 \text{ miles}}$ 2.8 days traveling along the river		Recommended Scoring (8 Total Points)	Consider giving partial credit for students who correctly show work.

Question	8	Benchmark(s)	MA.7.AR.3.2, MA.7.AR.4.5
$\frac{20 \text{ tsp}}{120 \text{ oz}} = \frac{x \text{ tsp of cocoa powder}}{16 \text{ oz}}$ $2\frac{2}{3} \text{ tsp}$		Recommended Scoring (8 Total Points)	Consider giving partial credit for students who correctly show work.
		Additional Notes	Students may choose to give the answer in an equivalent form. If the student responds using a decimal without the repeating bar consider deducting one point.
Question	9	Benchmark(s)	MA.7.AR.3.2, MA.7.AR.4.5
$\frac{2.25 \text{ in}}{180 \text{ in}^3} = \frac{3.5 \text{ in}}{x \text{ in}^3}$ $x = 280 \text{ in}^3$ $280 \text{ in}^3 \times 3 = 840 \text{ in}^3$		Recommended Scoring (8 Total Points)	Consider giving partial credit for students who correctly show work.
Question	10	Benchmark(s)	MA.7.AR.3.2, MA.7.AR.4.5
$\frac{4.4 \text{ cm}}{11.8 \text{ cm}} = \frac{x \text{ miles from Polk City Trail Head to the Withlacoochee River}}{29.2 \text{ miles}}$ 10.9 miles		Recommended Scoring (8 Total Points)	Consider giving partial credit for students who correctly show work.